Safe Bayesian Optimization Across Noise Models via Scenario Programming

Abdullah Tokmak, Thomas B. Schön, *Senior Member, IEEE*, and Dominik Baumann, *Member, IEEE*

***Abstract*—Safe Bayesian optimization (BO) with Gaussian processes is an effective tool for tuning control policies in safety-critical real-world systems, specifically due to its sample efficiency and safety guarantees. However, most safe BO algorithms assume homoscedastic sub-Gaussian measurement noise, an assumption that does not hold in many relevant applications. In this letter, we propose a straightforward yet rigorous approach for safe BO across noise models, including homoscedastic sub-Gaussian and heteroscedastic heavy-tailed distributions. We provide a high-probability bound on the measurement noise via the scenario approach, integrate these bounds into high probability confidence intervals, and prove safety and optimality for our proposed safe BO algorithm. We deploy our algorithm in synthetic examples and in tuning a controller for the Franka Emika manipulator in simulation.**

***Index Terms*—Statistical learning, hyper parameter optimization, Gaussian processes, robot control.**

I. Introduction

LEARNING-BASED methods, such as reinforcement learning (RL) [1], have been successfully used to obtain high-performing policies for nonlinear systems with unknown dynamics. However, to tune control policies of safety-critical real-world systems, two essential criteria must be met. First, we require sample efficiency, since obtaining a sample corresponds to conducting a real-world experiment. Second, we need safety guarantees to ensure secure operation of the system. Vanilla RL has inherent difficulties with both criteria. Bayesian optimization (BO) [2] with Gaussian process (GP) [3] surrogates offers a sample-efficient alternative that works effectively in, e.g., robotics applications [4]. Moreover, safe BO algorithms [5] also provide probabilistic safety guarantees. However, most safe BO works assume—and are thus only robust with respect to—homoscedastic R-sub-Gaussian measurement noise. This assumption is broken in many relevant scenarios, such as in the modeling of network delays [6], measurements from radar or LiDAR sensors [7], or parameter tuning on real-world systems, where the observation noise often exhibits strong dependence on the chosen parameter [8], [9]. In this letter, we overcome this bottleneck by proposing a practical, safe BO algorithm that is straightforwardly configurable across different noise models.

Received 11 September 2025; revised 7 November 2025; accepted 27 November 2025. Date of publication 18 December 2025; date of current version 5 January 2026. This work was supported in part by the *Kjell och Märta Beijer Foundation.* Recommended by Senior Editor K. Savla. *(Corresponding author: Abdullah Tokmak.)*

Abdullah Tokmak and Dominik Baumann are with the Cyber-Physical Systems Group, Aalto University, 02150 Espoo, Finland (e-mail: abdullah.tokmak@aalto.fi; dominik.baumann@aalto.fi).

Thomas B. Schön is with the Department of Information Technology, Uppsala University, 751 05 Uppsala, Sweden (e-mail: thomas.schon@uu.se).

Digital Object Identifier 10.1109/LCSYS.2025.3645824

Related work. SafeOpt [5]—which extends GP-UCB [10], [11] to probabilistic constraint satisfaction—is arguably the most popular safe BO algorithm. SafeOpt and extensions have been applied to control parameter tuning for medical devices [5], quadrotor vehicles [12], the Franka Emika manipulator [13], and the Furuta pendulum [14]. Although the seminal SafeOpt work [5] requires independent and identically distributed (i.i.d.) Gaussian noise ϵ_t, [12], [13], [14], [15] relax this assumption to ϵ_t only being homoscedastic R-sub-Gaussian conditioned on the filtration, i.e., for all $\lambda \in \mathbb{R}$, $\mathbb{E}[\exp(\lambda\epsilon_t)|\mathcal{F}_{t-1}] \leq \exp(\lambda^2 R^2/2)$ for all iterations $t \geq 1$. In the sub-Gaussian case, the safe BO algorithms use confidence intervals derived in, e.g., [11], [16].

Nevertheless, many relevant applications [6], [7] are endowed by heavy-tailed distributions whose tails do not decay sufficiently fast to be classified as sub-Gaussian. In the linear bandit setting, [17], [18] propose algorithms capable of handling heavy-tailed noise, while [19] extends these approaches to the kernelized setting. These works only require a (known) bound on the $(1+\alpha)$-th moment for $\alpha \in (0, 1]$. Moreover, [20] proposes a BO algorithm based on GP-UCB for heavy-tailed rewards. However, the mentioned works [17], [18], [19], [20] only consider homoscedastic noise and do not account for constraints.

References [8], [9] argue that many use cases involve heteroscedastic noise, i.e., noise that depends on the input. To address such cases, [8] models the stochastic bandit problem under heteroscedastic sub-Gaussian noise and derives corresponding confidence intervals for functions in reproducing kernel Hilbert spaces (RKHSs). This idea is then extended in [9] to the risk-averse BO setting.

While the contributions mentioned above tackle specific noise assumptions, none of these works are applied to safe BO or account for heteroscedastic heavy-tailed distributions. More crucially, each noise assumption relies on its own theoretical foundation and demands intricate transformations for implementation. Altogether, there is no unified and practical yet rigorous framework for safe BO that is straightforwardly adjustable across noise models, including homoscedastic sub-Gaussian and heteroscedastic heavy-tailed distributions.

Contribution. We propose a safe BO algorithm reminiscent of SafeOpt that is straightforwardly configurable

2475-1456 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on November 01,2026 at 09:53:32 UTC from IEEE Xplore. Restrictions apply.

across different noise models. We make the following contributions:[1]

(C1) We propose a scenario-based approach [21] to obtain probabilistic noise bounds.
(C2) Using the bounds from (C1), we derive high probability confidence intervals.
(C3) We develop a safe BO algorithm reminiscent of SAFEOPT [5] using the confidence intervals from (C2), for which we prove safety and optimality.

II. Problem Setting and Preliminaries

Next, we explain the optimization problem (Section II-A), SAFEOPT (Section II-B), and the scenario approach (Section II-C).

A. Optimization Problem

We seek to maximize the unknown reward function $f : \mathcal{A} \subseteq \mathbb{R}^n \to \mathbb{R}$ subject to unknown constraints $g_i : \mathcal{A} \subseteq \mathbb{R}^n \to \mathbb{R}, i \in \mathcal{I}_g \subseteq \mathbb{N}$. Analogous to [12], we introduce a surrogate function h_i with $h_i(\cdot) = f(\cdot)$ if $i = 0$ and $h_i(\cdot) = g_i(\cdot)$ if $i \in \mathcal{I}_g$, and define $\mathcal{I} \coloneqq \{0\} \cup \mathcal{I}_g$. We assume that at each iteration $t \geq 1$, an experiment following a parametrized policy with policy parameter $a_t \in \mathcal{A}$ yields a noisy evaluation $y_{i,t} = h_i(a_t) + \epsilon_{i,t}$, and denote the observation noise vector by $\epsilon_t \coloneqq [\epsilon_{0,t}, \ldots, \epsilon_{|\mathcal{I}_g|,t}]^\top$.

Assumption 1: The observation noise ϵ_t is defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, from which we can sample.

Assumption 1 introduces a fairly general and unified approach for any noise model, as long as we can sample from its probability space. In contrast to the classic R-sub-Gaussian assumption [11], we assume that the noise can be sampled directly, which is not possible under the sub-Gaussian formulation since it represents a family of distributions rather than a specific one. This assumption enables computing high-probability noise bounds, as established later in Theorem 1. Arguably, the tail behavior of the noise is the most important aspect for achieving such bounds. Since the normal distribution $\mathcal{N}(0, R^2)$ saturates the moment-generating-function of the R-sub-Gaussian family and therefore exhibits the heaviest admissible tails, it provides an empirical and practically convenient conservative surrogate. Assumption 1 thereby unifies different noise models under a single theory, allowing practitioners to incorporate prior knowledge about the noise or to employ data-driven or oracle-based representations of the aleatoric uncertainty.

Altogether, the optimization problem is described by

$$\max_{a \in \mathcal{A}} f(a) \text{ s.t. } g_i(a_t) \geq 0, \quad \forall t \geq 1, \forall i \in \mathcal{I}_g, \tag{1}$$

i.e., we sample h episodically to maximize f while requiring safety. In (1), safety is defined as only conducting experiments that correspond to positive constraint values. We solve (1) using safe BO to exploit its sample efficiency and safety guarantees by making a smoothness assumption, which is standard in many BO algorithms, see, e.g., [5], [10], [11], [12], [13].

Assumption 2: For any $i \in \mathcal{I}$, h_i is a member of the RKHS of the continuous kernel k with known RKHS norm $\|h_i\|_k$.

[1]Code/experiments: https://github.com/tokmaka1/ACC-2026.

Note that, for brevity, we assume $k \equiv k_i, \forall i \in I$. We introduce the kernel metric $d_k(a, a') \coloneqq \sqrt{2(k(a,a) - k(a,a'))}$ to leverage a well-known result on the continuity of RKHS functions, which will assist us in formulating our safe BO algorithm.

Lemma 1 (Continuity [22, Lemma 1]): Under Assumption 2, $|h_i(a) - h_i(a')| \leq \|h_i\|_k d_k(a, a'), \ \forall a, a' \in \mathcal{A}, i \in \mathcal{I}$.

B. SafeOpt

Akin to SAFEOPT [12], we use independent GP surrogates with mean function $\mu_{i,t}$ and variance σ_t constructed using kernel k to model the unknown functions h_i. To guarantee safety while maximizing the reward, i.e., to solve (1), we must quantify the uncertainty around our belief of the ground truth h_i. Hence, we require high probability frequentist confidence intervals, as derived in [11], [16] for homoscedastic sub-Gaussian measurement noise, of the form

$$C_{i,t}(a) \coloneqq C_{i,t-1}(a) \cap \left[\mu_{i,t}(a) \pm \beta_{i,t}\sigma_t(a)\right], \tag{2}$$

for all $t \geq 1, i \in \mathcal{I}, a \in \mathcal{A}$, with $C_{i,0}(a) \coloneqq (-\infty, \infty)$ and a safety parameter $\beta_{i,t}$, which we will later make precise as we derive our bounds. Specifically, we will present confidence intervals that account for general noise models under Assumption 1. We introduce the upper and lower confidence bounds as $u_{i,t}(a) \coloneqq \max C_{i,t}(a)$ and $\ell_{i,t}(a) \coloneqq \min C_{i,t}(a)$, respectively, denote the uncertainty width by $w_{i,t}(a) \coloneqq u_{i,t}(a) - \ell_{i,t}(a)$, and define a safe set

$$S_t \coloneqq \cap_{i \in \mathcal{I}_g} \cup_{a \in S_{t-1}} \{a' \in \mathcal{A} | \ell_{i,t}(a) - \|h_i\|_k d_k(a, a') \geq 0\}. \tag{3}$$

The safe set only contains parameters that satisfy the constraints in (1) with high probability [5, Th. 1]. To start exploration, we assume that $\emptyset \neq S_0 \subseteq \mathcal{A}$, i.e., a non-empty set of initial safe samples is given. To efficiently balance exploration and exploitation while guaranteeing safety, we introduce the set of potential maximizers $M_t \coloneqq \{a \in S_t | u_{t,0} \geq \max_{a' \in S_t} \ell_{t,0}(a')\}$ and the set of potential expanders $G_t \coloneqq \{a \in S_t | e_t(a) > 0\}, e_t(a) = |\{b \in \mathcal{A} \setminus S_t | \exists i \in \mathcal{I}_g, u_{i,t}(a) - \|g_i\|_k d_k(a, b) \geq 0\}|$. The set M_t contains samples that we believe to safely maximize the reward, while an evaluation of the samples within G_t may safely expand the safe set. Akin to SAFEOPT, the acquisition function is $a_{t+1} = \arg\max_{a_i \in (M_t \cup G_t)} \max_{i \in \mathcal{I}} w_{i,t}(a_i)$, i.e., we conduct the next experiment with the most uncertain parameter in $M_t \cup G_t$.

C. Scenario Approach

Throughout this letter, we derive confidence intervals $C_{i,t}$ (see (2)) that are adjustable across different noise models by leveraging Assumption 1 and the scenario approach [21].

Let us denote by $\tilde{\epsilon}_t^{(j)}, j \in [1, m_t]$, the *scenarios*, which are drawn i.i.d. from the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Using the scenarios $\tilde{\epsilon}_t^{(j)}$, we aim to obtain a high probability scenario bound $\bar{\epsilon}_t \coloneqq [\bar{\epsilon}_{0,t}, \ldots, \bar{\epsilon}_{|\mathcal{I}_g|,t}]^\top$ on the unknown observation noise ϵ_t by solving the convex scenario program

$$\min_{\bar{\epsilon}_t \in \mathbb{R}^{|\mathcal{I}|}_{\geq 0}} \mathbf{1}^\top \bar{\epsilon}_t \quad \text{s.t.} \quad \bar{\epsilon}_{i,t} \geq |\tilde{\epsilon}_{i,t}^{(j)}|, \quad \forall j \in [1, m_t], \tag{4}$$

where $\mathbf{1} \coloneqq [1, \ldots, 1]^\top$. Since we minimize $\bar{\epsilon}_t$ element-wise,

$$\bar{\epsilon}_t = \left[\max_{j \in [1,m_t]} |\tilde{\epsilon}_{0,t}^{(j)}|, \ldots, \max_{j \in [1,m_t]} |\tilde{\epsilon}_{|\mathcal{I}_g|,t}^{(j)}|\right]^\top \tag{5}$$

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on November 01,2026 at 09:53:32 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Bounding General Noise

Require: κ, ν, $(\Omega, \mathcal{F}, \mathbb{P})$, t, $|\mathcal{I}|$
1: $\kappa_t \leftarrow 6\kappa/\pi^2 t^2$
2: $m_t \leftarrow \min_{m_t \in \mathbb{N}}$ s.t. $\sum_{s=0}^{|\mathcal{I}|-1} \binom{m_t}{s} \nu^s (1-\nu)^{m_t - s} \leq \kappa_t$
3: Generate m_t i.i.d. scenarios ϵ_j from $(\Omega, \mathcal{F}, \mathbb{P})$
4: **return** $\bar{\epsilon}_t := [\bar{\epsilon}_{0,t}, \ldots, \bar{\epsilon}_{|\mathcal{I}_g|,t}]^\top$ ▷ (5)

is the solution to (4). Remarkably, by solving (4), i.e., by generating a finite number of scenarios, the scenario approach generalizes to the true random variable ϵ_t as long as it lives on the same probability space. To quantify the generalization property, we define the violation probability as under-estimating the absolute value of the observation noise, i.e.,

$$V_t(\bar{\epsilon}_t) := \mathbb{P}\big[\exists i \in \mathcal{I} : \bar{\epsilon}_{i,t} < |\epsilon_{i,t}|\big]. \tag{6}$$

The following lemma is a classic result from scenario theory and bounds the violation probability $V_t(\bar{\epsilon}_t)$.

Lemma 2 (Scenario Approach [21, Th. 3.7]): Fix any $t \geq 1$, let Assumption 1 hold, and $\tilde{\epsilon}_t^{(j)}$, $j \in [1, m_t]$ be i.i.d. scenarios from $(\Omega, \mathcal{F}, \mathbb{P})$. Then, for any $\nu, \kappa \in (0,1)$, if m_t is such that $\sum_{s=0}^{|\mathcal{I}|-1} \binom{m_t}{s} \nu^s (1-\nu)^{m_t - s} \leq \kappa$, then $\mathbb{P}^{m_t}[V_t(\bar{\epsilon}_t) > \nu] \leq \kappa$, with $\bar{\epsilon}_t$ as in (5).

Lemma 2 states for any fixed iteration $t \geq 1$ that the violation probability (6) is larger than ν with confidence of at most κ. Specifically, it quantifies the probability in the product probability space $(\Omega^{m_t}, \mathcal{F}^{m_t}, \mathbb{P}^{m_t})$, which naturally arises from the m_t scenarios being i.i.d. samples from $(\Omega, \mathcal{F}, \mathbb{P})$.

III. Framework for Bounding General Noise

We derive probabilistic bounds on the measurement noise that hold simultaneously for all iterations using the scenario approach (Sec. III-A) before constructing confidence intervals that are adjustable across different noise models (Section III-B).

A. Scenario-Based Simultaneous Bounds

Although Lemma 2 provides a scenario-based high-probability noise bound, it only captures a *fixed* time step t. To provide guarantees on the measurement noise while solving (1), we require probabilistic bounds that hold *simultaneously* for all $t \geq 1$. To this end, we present Algorithm 1.

Specifically, we want the violation probability (6) to be bounded by a user-chosen confidence level κ simultaneously for all iterations t. Algorithm 1 computes the number of scenarios m_t (ℓ. 2) based on an iteration-adjusted confidence level κ_t (ℓ. 1). Then, it generates the scenarios (ℓ. 3) and returns the bound $\bar{\epsilon}_t$ (5), which solves the scenario program (4). In Theorem 1, we will prove that satisfying the iteration-adjusted confidence level κ_t at each iteration indeed provides a scenario bound $\bar{\epsilon}_t$ that bounds (6) with probability ν simultaneously for all t with the desired confidence level κ.

We proceed by formulating the probability space in which we define our guarantees. For Lemma 2, we already constructed a product probability space over the m_t i.i.d. sampled constraints. By sampling the constraints independently over the number of iterations, we can extend the product probability space $(\Omega^{m_t}, \mathcal{F}^m, \mathbb{P}^{m_t})$ to a product probability space that is valid across all iterations $t \geq 1$, which we refer to as the master probability space. Due to the independence across the iterations, the master probability space can be written as $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathbb{P}})$, with $\tilde{\Omega} := \otimes_{t=1}^{\infty} \Omega^{m_t}$, $\tilde{\mathcal{F}} := \otimes_{t=1}^{\infty} \mathcal{F}^{m_t}$, $\tilde{\mathbb{P}} := \otimes_{t=1}^{\infty} \mathbb{P}^{m_t}$. We are now ready to lift Lemma 2 to hold for all iterations simultaneously.

Theorem 1 (Simultaneous Scenario Bound): Let Assumption 1 hold, $\tilde{\epsilon}_t^{(j)}, j \in [1, m_t]$ be i.i.d. samples from $(\Omega, \mathcal{F}, \mathbb{P})$, and receive $\bar{\epsilon}_t$ from Algorithm 1 for all $t \geq 1$. Then, for any $\gamma, \kappa \in (0,1)$, the violation probability (6) is bounded by $\tilde{\mathbb{P}}[V_t(\bar{\epsilon}_t) > \nu] \leq \kappa$ *simultaneously* for all $t \geq 1$.

Proof: The guarantees in this theorem are given with respect to the master probability measure $\tilde{\mathbb{P}}$. Since $\mathbb{P}^m$ can be seen as a projection of the master probability measure $\tilde{\mathbb{P}}$, we can write the results from Lemma 2 with respect to $\tilde{\mathbb{P}}$ by fixing an iteration t. Therefore, $\tilde{\mathbb{P}}[V_t(\bar{\epsilon}_t) > \nu] \leq \sum_{s=0}^{|\mathcal{I}|-1} \binom{m_t}{s} \nu^s (1-\nu)^{m_t - s}$ for any (fixed) t. To lift this result to all iterations, we first note that the outer probability measure (in this case $\tilde{\mathbb{P}}$) quantifies the uncertainty over time—the inner probability space governed by the measure $\mathbb{P}$ does not depend on time since it is fully characterized by $\tilde{\epsilon}_t^{(j)}$. Therefore, the simultaneous bound is $\tilde{\mathbb{P}}[\forall t \geq 1 : V_t(\bar{\epsilon}_t) > \nu] = \tilde{\mathbb{P}}[\cup_{t=1}^{\infty} V_t(\bar{\epsilon}_t) > \nu] \leq \sum_{t=1}^{\infty} \tilde{\mathbb{P}}[V_t(\bar{\epsilon}_t) > \nu]$, by Boole's inequality. Algorithm 1 computes the number of m_t (ℓ. 2) such that $\bar{\epsilon}_t$ (ℓ. 4) yields a ν-bounded violation probability (6) with the iteration-adjusted confidence level κ_t, i.e., $\tilde{\mathbb{P}}[V_t(\bar{\epsilon}_t) > \nu] \leq \kappa_t$, see Lemma 2. Altogether, we have $\tilde{\mathbb{P}}[\forall t \geq 1 : V_t(\bar{\epsilon}_t) > \nu] \leq \sum_{t=1}^{\infty} \kappa_t \leq \kappa$, since $\kappa_t := 6\kappa/\pi^2 t^2$ (ℓ. 1) and $\sum_{t=1}^{\infty} \frac{1}{t^2} = \frac{\pi^2}{6}$. ■

Theorem 1 establishes that with confidence $1-\kappa$, the scenario-based bounds computed by Algorithm 1 hold uniformly across all iterations with probability $1-\nu$ (C1).

B. High Probability Confidence Intervals

Let us now present confidence intervals by leveraging the scenario-based noise bound derived in Theorem 1 (C2).

Corollary 1 (Confidence Intervals): Let the conditions in Theorem 1 and Assumption 2 hold. For any $\gamma, \kappa \in (0,1)$ and any fixed regularization constant of the GP posterior $\eta \in (0,1]$,

$$\tilde{\mathbb{P}}\big[\forall t \geq 1, i \in \mathcal{I}, a \in \mathcal{A} : \mathbb{P}[h_i(a) \in C_{i,t}] \geq 1 - \nu\big] \geq 1 - \kappa,$$

with $C_{i,t}$ as defined in (2) and

$$\beta_{i,t} := \|h_i\|_k + \sqrt{\lambda_{\max}(\Xi_t)/\eta}\,\|\bar{\epsilon}_{i,1:t}\|_2, \tag{7}$$

where $K_t \in \mathbb{R}^{t \times t}$ is the covariance matrix, $\Xi_t := K_t(K_t + \eta I_t)^{-1}$, and $\lambda_{\max}(\Xi_t)$ denotes the largest eigenvalue of Ξ_t.

Proof: We follow the proof of [11, Th. 2], adjusted to regularization parameters $\eta > 0$ as in [16, Th. 1]. However, instead of bounding the norm of the noise vector $\epsilon_{i,1:t}$ using a surrogate supermartingale, we use our result developed in Theorem 1. The proofs in [11], [16] establish that $|h_i(a) - \mu_{i,t}(a)| \leq (\|h_i\|_k + \sqrt{1/\eta}\|\epsilon_{i,1:t}\|_{\Xi_t})\sigma_t(a)$ holds deterministically, with $\|\epsilon_{i,1:t}\|_{\Xi_t} := \epsilon_{i,1:t}^\top \Xi_t \epsilon_{i,1:t}$. For the noise norm, we have that $\|\epsilon_{i,1:t}\|_{\Xi_t} \leq \sqrt{\lambda_{\max}(\Xi_t)}\|\epsilon_{i,1:t}\|_2$ by the Rayleigh–Ritz theorem since Ξ_t is Hermitian. Moreover, Theorem 1 ensures $\bar{\epsilon}_{i,t} \geq |\epsilon_{i,t}|$ for all $i \in \mathcal{I}, t \geq 1$ with high probability. Since $\bar{\epsilon}_{i,t}$ is non-negative, this implies $\|\bar{\epsilon}_{i,1:t}\|_2 \geq \|\epsilon_{i,1:t}\|_2$ with high probability. Altogether, $\tilde{\mathbb{P}}[\forall t \geq 1, \forall i \in \mathcal{I} : \mathbb{P}[\sqrt{\lambda_{\max}(\Xi_t)}\|\bar{\epsilon}_{i,1:t}\|_2 < \|\epsilon_{i,1:t}\|_{\Xi_t}] > \nu] \leq \kappa$, which

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on November 01,2026 at 09:53:32 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2 Safe BO Under General Noise

Require: k, $\mathcal{A}$, S_0, $\|h_i\|_k$, η, δ, $\mathcal{I}$,
1: **for** $t = 1, 2, \ldots$ **do**
2: Compute GP posterior mean $\mu_{i,t}$ and variance σ_t
3: $\bar{\epsilon}_t \leftarrow$ Algorithm 1
4: Build confidence intervals ▷ Corollary 1, (2)
5: **if** $t > 1$ **then** $S_t \leftarrow$ (3) **else** $S_t \leftarrow S_0$
6: Compute $w_{i,t}$ and sets M_t, G_t
7: $a_{t+1} = \arg\max_{a_i \in (M_t \cup G_t)} \max_{i \in \mathcal{I}} w_{i,t}(a_i)$
8: **if** $w_{i,t}(a_{t+1}) < \delta$ **then** break
9: $y_{i,t+1} \leftarrow h_i(a_{t+1}) + \epsilon_{i,t}$ ▷ Experiment
10: **return** $\arg\max_{a \in S_t} \ell_{0,t}(a)$ ▷ Optimal parameter

implies $\tilde{\mathbb{P}}[\forall t \geq 1, i \in \mathcal{I}, a \in \mathcal{A} : \mathbb{P}[h_i(a) \in [\mu_{i,t}(a) \pm \beta_{i,t}\sigma_t(a)]] \geq 1 - \nu] \geq 1 - \kappa$, with $\beta_{i,t}$ as in (7). ■

Note that the nature of the noise enters in Theorem 1 when generating scenarios from the noise's probability distribution.

IV. Safe BO Across Noise Models

In Section IV-A, we present our safe BO algorithm that accommodates observation noise under various noise models. We prove safety and optimality of our algorithm in Section IV-B.

A. Algorithm

Algorithm 2 builds up on SafeOpt and summarizes our safe BO algorithm to solve (1). First, we compute the GP mean and variance (ℓ. 2) before receiving the scenario bound $\bar{\epsilon}_t$ from Algorithm 1 (ℓ. 3). After building the confidence intervals (ℓ. 4), we compute the required sets (ℓ. 5-6) before acquiring the next parameter a_{t+1} (ℓ. 7). If the uncertainty of a_{t+1} is larger than the user-defined exploration threshold $\delta > 0$ (ℓ. 8), we conduct an experiment (ℓ. 9) and continue the procedure. Otherwise, we return the parameter that we estimate to safely maximize the reward (ℓ. 10).

B. Safety and Optimality

Next, we prove safety and optimality of Algorithm 2. Since exploration is restricted to the safe set $S_t \subseteq \mathcal{A}$, we can only hope to identify the maximizer of the reward h_0 within a subset of the domain $\mathcal{A}$. Thus, equivalent to [5], [12], we introduce the reachability operator $R_\delta(S) := S \cup \cap_{i \in \mathcal{I}_g}\{a \in \mathcal{A} | \exists a' \in S : h_i(a') - \delta - \|h_i\|_k d_k(a, a') \geq 0\}$. Moreover, we denote the reachable set by $\bar{R}_\delta(S) := \lim_{s \to \infty} R_\delta^s(S)$ with $R_\delta^s(S) := R_\delta(R_\delta^{s-1}(S))$. Before proving that we safely converge to the δ-reachable maximizer within $\bar{R}_\delta(S)$, we introduce the maximum information gain $\gamma_{|\mathcal{I}|t}$, which bounds the information we can obtain about h_i from measurements; see [10], [12] for details. Note that obtaining $\gamma_{|\mathcal{I}|t}$ is non-trivial. Nevertheless, unlike [5], [10], [12], we do not require $\gamma_{|\mathcal{I}|t}$ to execute Algorithm 2.

Theorem 2 (Safety and Optimality): Let the conditions in Corollary 1 hold, a non-empty initial safe set of parameters S_0 be given, and define $a_t^\star := \arg\max_{a \in S_t} \ell_{t,0}(a)$ with

$$t^\star \geq \left\lceil \frac{8\bar{\beta}_{t^\star}^2 \gamma_{|\mathcal{I}|t^\star}\left(|\bar{R}_0(S_0)| + 1\right)}{\log\left(1 + \eta^{-2}\right)\delta^2} \right\rceil,$$

where $\lceil \zeta \rceil$ denotes rounding $\zeta \in \mathbb{R}$ up to the next integer and $\bar{\beta}_t := \max_{i \in \mathcal{I}} \beta_{i,t}$. Then, for any $\nu, \kappa, \delta \in (0, 1)$ and any $\eta \in (0, 1]$, Algorithm 2 terminates after at most $t^\star$ iterations, and the following two statements hold simultaneously with probability at least $1 - \nu$ and confidence at least $1 - \kappa$:

- *Safety:* $h_i(a_t) \geq 0, \forall i \in \mathcal{I}_g, \forall t \in [1, t^\star]$;
- *δ-reachable optimality:* $\ell_{0,t^\star}(a_t^\star) \geq \max_{a \in \bar{R}_\delta(S_0)} h_0(a) - \delta$.

Proof: To guarantee safety, [5, Lemma 11] and [12, Lemma 7.9] prove that $\forall a \in S_t, i \in, t \geq 1, i \in \mathcal{I}_g : h_i(a) \geq 0$ with high probability, while [14, Lemma 2] extends this result to unknown Lipschitz constants by exploiting Lemma 1. We adjust [14, Lemma 2] using our confidence intervals derived in Corollary 1 to obtain $\tilde{\mathbb{P}}[\forall t \geq 1, \forall i \in \mathcal{I}_g, \forall a \in S_t : \mathbb{P}[h_i(a) \geq 0] \geq 1 - \nu] \geq 1 - \kappa$, from which safety directly follows since Algorithm 2 only samples within S_t. The optimality proof proceeds analogously to the proofs of [5, Th. 1] and [12, Th. 4.1]. The key differences from their proofs are that we rely on the RKHS norm instead of Lipschitz constant for continuity (Lemma 1) and that we employ the confidence intervals from Corollary 1, which yield guarantees holding with probability $1 - \nu$ and confidence $1 - \kappa$. Also, we do not ensure that $\bar{\beta}_t$ is non-decreasing in t by exploiting the monotonicity of the information gain. Instead, $\bar{\beta}_t$—and $\beta_{i,t} \forall i \in \mathcal{I}$ (7)—are non-decreasing in t since $\lambda_{\max}(\Xi_t) = \lambda_{\max}(K_t(K_t + \eta I_t)^{-1}) = \frac{\lambda_{\max}(K_t)}{\lambda_{\max}(K_t) + \eta}$, and $\lambda_{\max}(K_t)$ is non-decreasing in t due to Cauchy's eigenvalue interlacing theorem. ■

Theorem 2 states that Algorithm 2 remains safe and terminates (see ℓ. 8) after at most $t^\star$ iterations, i.e., when it attains δ-reachable optimality (C3). We not only accommodate observation noise across various noise models, but also ensure that the theoretical guarantees of Theorem 2 hold in practice. This follows from working directly with the safe set as defined in (3), while [12, Sec. 4.3] instead employs an alternative approach that avoids the Lipschitz constant but does so at the expense of exploration guarantees. In fact, most BO algorithms [10], [11], [12], [13] fix a scalar value for $\beta_{i,t}$, which makes their safety and optimality guarantees mundane in practice.

Remark 1 (Finite-Time Convergence): For $t^\star < \infty$, we require $\bar{\beta}_t \leq \mathcal{O}(\sqrt{t})$; a bound on the growth rate that we do not establish. In fact, adversarial noise models can be chosen that ensure $\bar{\beta}_t > \mathcal{O}(\sqrt{t})$. We do not characterize the exact conditions under which finite-time convergence is guaranteed and only examine the growth rate of $\bar{\beta}_t$ only for a specific case later in Section V-B (Figure 5).

V. Experiments

In Section V-A, we evaluate Algorithm 2 using synthetic experiments with homoscedastic sub-Gaussian and heteroscedastic heavy-tailed noise. In Section V-B, Algorithm 2 safely tunes a controller for the Franka Emika robot, while Section V-C discusses the computational complexity of the scenario-based bounds. All experiments were conducted with hyperparameters $\nu = 10^{-1}$, $\kappa = 10^{-3}$ and the Matérn32 kernel with lengthscale $\ell = 0.1$. We compute $\beta_{i,t}$ as in (7) with $\|h_i\|_k = 1$.

A. Synthetic Examples

We quantitatively compare Algorithm 2 only with the standard homoscedastic sub-Gaussian safe BO bounds [11], as our algorithm can operate under this standard setting while

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on November 01,2026 at 09:53:32 UTC from IEEE Xplore. Restrictions apply.

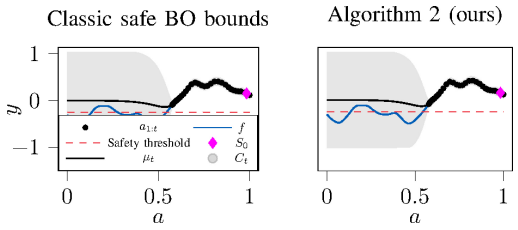


Fig. 1. *Homoscedastic sub-Gaussian noise:* Algorithm 2 (ours, right) safely explores the domain to identify the maximizer with the confidence intervals specified in Corollary 1. The performance is comparable to classic safe BO algorithms (left).

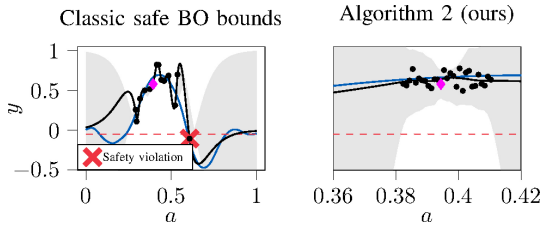


Fig. 2. *Heteroscedastic heavy-tailed noise:* The right sub-figure provides a magnified view. Algorithm 2 (right, ours) explores carefully and remains safe since the confidence intervals from Corollary 1 can account for heteroscedastic heavy-tailed noise. In contrast, classic safe BO algorithms (left) explore more optimistically but do not account for this noise type, which yields safety violations (red cross).

generalizing straightforwardly to heteroscedastic and heavy-tailed noise models through appropriate scenario generation.

Homoscedastic sub-Gaussian noise. We consider a scalar function $h_i \equiv f$, i.e., we have $|\mathcal{I}| = 1$. The reward function f is a random function from the pre-RKHS of kernel k, whose coefficients we scale to obtain $\|f\|_k = 1$; see [16, Appendix C.1] for details. We generate the observation noise from a homoscedastic uniform distribution with $\epsilon_t \sim \mathcal{U}(-10^{-3}, 10^{-3})$, scale the safety threshold to be the 40% quantile of f, and use the regularization factor $\eta = 10^{-2}$ and exploration threshold $\delta = 10^{-1}$. Figure 1 illustrates the performance of our safe BO algorithm with confidence intervals derived in Corollary 1 (right) and standard safe BO algorithms (left). Algorithm 2 safely explores the domain and identifies the maximizer starting from the initial safe set S_0. The performance is comparable to classic safe BO algorithms that work with sub-Gaussian confidence intervals derived in [11], [16] with $R = 10^{-3}$. Next, we show that working with classic safe BO bounds outside of this noise assumption may yield safety violations.

Heteroscedastic heavy-tailed noise. We examine an equivalent setting as above. However, we here consider heteroscedastic and heavy-tailed observation noise $\epsilon_t(a) \sim |a|/5 \cdot \mathcal{T}_{10}$, where $\mathcal{T}_{10}$ denotes the Student's-t distribution with ten degrees of freedom. We further set $R = 10^{-5}$ and $\eta = 10^{-3}$. Figure 2 compares Algorithm 2 (right) with confidence intervals from Corollary 1 to classic safe BO algorithms (left) with R-sub-Gaussian confidence intervals. By leveraging scenario-based bounds on the observation noise, Algorithm 2 accommodates heteroscedastic heavy-tailed behavior, enabling more cautious exploration while maintaining safety. In contrast, classical safe BO algorithms tend to explore overly optimistically, which can ultimately lead to safety violations.

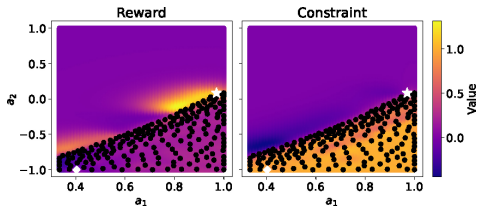


Fig. 3. *Franka experiment:* Exploration behavior and normalized GP mean values of the reward (left) and the constraint (right).

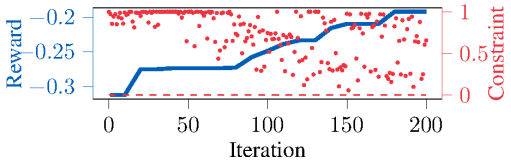


Fig. 4. *Franka experiment:* The maximum reward (left y-axis, blue) is increasing while the constraint value (right y-axis, red) remains above the safety threshold (dashed red line).

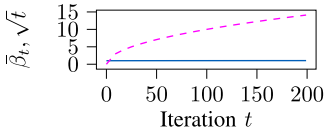


Fig. 5. Growth of confidence parameter $\bar{\beta}_t$ (blue) and $\sqrt{t}$ (magenta) over iterations t.

B. Control Parameter Tuning

We next consider a set-point tracking task for the Franka Emika robot in a simulation environment based on Mujoco [23]. Our setting is akin to [13, Sec. 5.1.1]. We utilize the same operational space impedance controller with impedance gain K and follow a linear quadratic regulator (LQR) type approach. That is, we define the diagonal matrices Q and R as $Q = \mathrm{diag}(Q_\mathrm{r}, \kappa_\mathrm{d} Q_\mathrm{r})$, where $Q_\mathrm{r} = 10^{q_\mathrm{c}} I_3$, and $R = 10^{r-2} I_3$. We obtain the gain K by solving the LQR problem. Here, we heuristically set $\kappa_\mathrm{d} = 0.1$ and are then left with $q_\mathrm{c} \in [1/3, 1]$ and $r \in [-1, 1]$ as the tuning parameters. The reward function f encourages reaching the target quickly with small inputs, whereas the constraint function g_1 requires that the distance to the target decreases sufficiently.

We compute the confidence intervals (Corollary 1) assuming that the inherent measurement noise follows a homoscedastic normal distribution with $\epsilon_{i,t} \sim \mathcal{N}(0, 10^{-4})$. We further set $\delta = 10^{-3}$, $\eta = 10^{-2}$ and start with the safe set $S_0 = \{[0.39, -1]^\top\}$. Figure 3 depicts the exploration behavior and the normalized GP means of the reward $\mu_{0,200}$ and the constraint $\mu_{1,200}$ after tuning the parameters for 200 iterations. Clearly, the initial value (white diamond) yields a smaller reward than the optimal parameter after 200 iterations (white star). The optimization task is particularly challenging, as the region of higher rewards coincides with low constraint values. Further, Figure 4 illustrates that no experiment incurred a safety violation, i.e., no experiment yielded a negative constraint value, and that the return of the currently believed optimum is increasing.

Moreover, we numerically examine the growth of $\bar{\beta}_t$ in Figure 5 and observe $\bar{\beta}_t \leq \mathcal{O}(\sqrt{t})$—and hence $t^\star < \infty$ in

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on November 01,2026 at 09:53:32 UTC from IEEE Xplore. Restrictions apply.

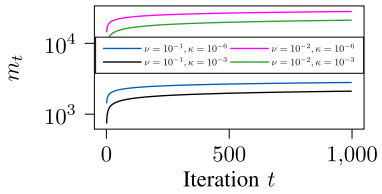


Fig. 6. Scaling of the number of scenarios m_t in Algorithm 1 over the number of iterations t. We consider $|\mathcal{I}| = 1$.

Theorem 2—for this particular experiment, although a formal guarantee for this behavior in general settings is not provided in this letter; see Remark 1.

C. Computational Complexity of Scenario-Based Bounds

Next, we remark on the computational complexity of the scenario-based bounds we use in Algorithm 1. First, we examine the effects of the user-chosen accuracy parameters $\nu, \kappa \in (0, 1)$. Figure 6 shows that the required number of scenarios m_t scales *linearly* with ν. In contrast, m_t only scales *logarithmically* with κ. In fact, the logarithmic scaling of m_t with κ is a well-known result of the scenario approach [21], which ensures that even extremely small confidence levels κ have only a mild effect on m_t. Furthermore, Figure 6 also depicts the effect of the iteration counter t on m_t. This dependence arises due to Algorithm 1 requiring an iteration-dependent confidence level κ_t (Algorithm 1, ℓ. 1-2). As t increases, the corresponding decrease in κ_t induces only a benign growth of m_t, consistent with our earlier remark on its merely logarithmic dependence on the confidence level. Lastly, generating scenarios $\tilde{\epsilon}_t^{(j)}$ in practice is usually computationally cheap if the noise is sampled from natively implemented probability distributions in standard libraries. However, if we require importance sampling or Markov chain Monte Carlo methods, the scenario generation may become more complex.

Finally, note that we do not require real-time capabilities with respect to the sampling time of the closed-loop dynamics since we work in an *episodic* control parameter tuning setting.

VI. Conclusion

We presented a safe BO algorithm reminiscent of SafeOpt that is straightforwardly adjustable across noise models. We proposed high-probability observation noise bounds using the scenario approach (C1) and integrated these bounds into frequentist confidence intervals (C2). Using these confidence intervals, we proved safety and optimality of our safe BO algorithm (C3). Our safe BO algorithm safely explored and optimized reward functions in synthetic examples under homoscedastic sub-Gaussian and heteroscedastic heavy-tail measurement noise. We deployed our algorithm in tuning control parameters of safety-critical systems by tuning a controller for the Franka Emika manipulator in simulation. Future work includes exploring data-driven noise oracles rather than analytical noise models and investigating the growth rate of $\tilde{\beta}_t$.

References

[1] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*. Cambridge, MA, USA: MIT press, 2018.
[2] R. Garnett, *Bayesian Optimization*. Cambridge, U.K.: Cambridge Univ., 2023.
[3] C. E. Rasmussen and C. K. Williams, *Gaussian Processes for Machine Learning*. Cambridge, MA, USA: MIT press, 2006.
[4] R. Calandra, A. Seyfarth, J. Peters, and M. P. Deisenroth, "Bayesian optimization for learning gaits under uncertainty," *Ann. Math. Artif. Intell*, vol. 76, pp. 5–23, Feb. 2016.
[5] Y. Sui, A. Gotovos, J. Burdick, and A. Krause, "Safe exploration for optimization with Gaussian processes," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 997–1005.
[6] K. Jagannathan, M. G. Markakis, E. Modiano, and J. N. Tsitsiklis, "Throughput optimal scheduling over time-varying channels in the presence of heavy-tailed traffic," *IEEE Trans. Inf. Theory*, vol. 60, no. 5, pp. 2896–2909, May 2014.
[7] H. Zou, S. Wu, Q. Xue, X. Sun, and B. Wang, "VB-T PHD-SLAM: Efficient SLAM under heavy-tailed noise," *Adv. Robot.*, vol. 38, no. 15, pp. 1093–1105, 2024.
[8] J. Kirschner and A. Krause, "Information directed sampling and bandits with heteroscedastic noise," in *Proc. Conf. Learn. Theory*, 2018, pp. 1–27.
[9] A. Makarova, I. Usmanova, I. Bogunovic, and A. Krause, "Risk-averse heteroscedastic Bayesian optimization," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 17235–17245.
[10] N. Srinivas, A. Krause, S. M. Kakade, and M. W. Seeger, "Information-theoretic regret bounds for Gaussian process optimization in the bandit setting," *IEEE Trans. Inf. Theory*, vol. 58, no. 5, pp. 3250–3265, May 2012.
[11] S. R. Chowdhury and A. Gopalan, "On kernelized multi-armed bandits," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 1–10.
[12] F. Berkenkamp, A. Krause, and A. P. Schoellig, "Bayesian optimization with safety constraints: Safe and automatic parameter tuning in robotics," *Mach. Learn.*, vol. 112, pp. 3713–3747, Oct. 2023.
[13] B. Sukhija, M. Turchetta, D. Lindner, A. Krause, S. Trimpe, and D. Baumann, "GoSafeOpt: Scalable safe exploration for global optimization of dynamical systems," *Artif. Intell.*, vol. 320, Jul. 2023, Art. no. 103922.
[14] A. Tokmak, K. G. Krishnan, T. B. Schön, and D. Baumann, "Safe exploration in reproducing kernel Hilbert spaces," in *Proc. Int. Conf. Artif. Intell. Stat.*, 2025, pp. 1–30.
[15] A. Tokmak, T. B. Schön, and D. Baumann, "PACSBO: Probably approximately correct safe Bayesian optimization," in *Proc. Symp. Syst. Theory Data Optim.*, 2024, pp. 1–15.
[16] C. Fiedler, C. W. Scherer, and S. Trimpe, "Practical and rigorous uncertainty bounds for Gaussian process regression," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 1–9.
[17] S. Bubeck, N. Cesa-Bianchi, and G. Lugosi, "Bandits with heavy tail," *IEEE Trans. Inform. Theory*, vol. 59, no. 11, pp. 7711–7717, Nov. 2013.
[18] H. Shao, X. Yu, I. King, and M. R. Lyu, "Almost optimal algorithms for linear stochastic bandits with heavy-tailed payoffs," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 8430–8439.
[19] A. Tajdini, J. Scarlett, and K. Jamieson, "Improved regret bounds for linear bandits with heavy-tailed rewards," in *Proc. Adv. Neural Inf. Process. Syst.*, 2025, pp. 1–23.
[20] S. R. Chowdhury and A. Gopalan, "Bayesian optimization under heavy-tailed payoffs," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 1–12.
[21] M. C. Campi and S. Garatti, *Introduction to the Scenario Approach*. Philadelphia, PA, USA: Soc. Ind. Appl. Math., 2018.
[22] A. Tokmak, C. Fiedler, M. N. Zeilinger, S. Trimpe, and J. Köhler, "Automatic nonlinear MPC approximation with closed-loop guarantees," *IEEE Trans. Automat. Control*, to be published.
[23] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *Proc. Int. Conf. Intell. Robots Syst.*, 2012, pp. 5026–5033.

Authorized licensed use limited to: Technische Universitaet Muenchen. Downloaded on November 01,2026 at 09:53:32 UTC from IEEE Xplore. Restrictions apply.